

ECE 447

Lesson 20

PCM (cont'd) and Pulse Modulations



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW3.** Posted on course website. Due Lesson 20 (Fri, 2 Oct, TONIGHT). Not on Prog grade.
 - **GR1 Grades.** Added a 5pt "curve" to everyone's GR1 grade in Blackboard. Not going to offer regrade opportunity (class Prog average is actually 1% higher than last semester's). Make sure you know what you missed, so you don't potentially miss it again on the Final.
 - Everything for Prog is graded and in Blackboard
 - Check your grades for any mistakes
 - Submit any regrade requests before 1600, 1 Oct

PCM COMPUTER EXAMPLE

- What has a larger impact on quantization error: sampling frequency or number of bits?

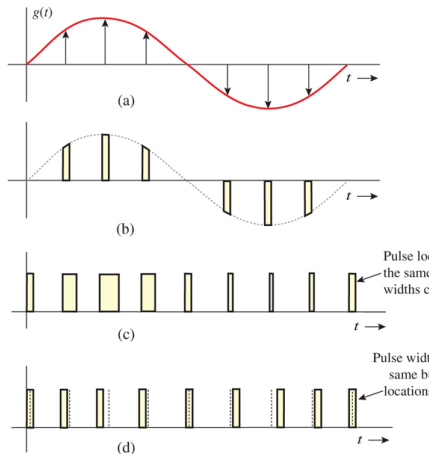
BINARY PCM

- SNR
 - Rule of thumb: each additional bit adds 6dB of SNR (how many times greater is that?)
- Max information rate given finite bandwidth
 - Assuming a noiseless channel with bandwidth B , a max of $2B$ pieces of info (e.g., bits) per second can be transmitted error-free
 - Shannon-Hartley theorem defines limit in presence of noise
- Transmission Bandwidth
 - For $m(t)$ band-limited to B Hz, min sample rate is $2B$ samples/s
 - Encoder output is then $2nB$ bit/s
 - Minimum required (theoretical) channel bandwidth,

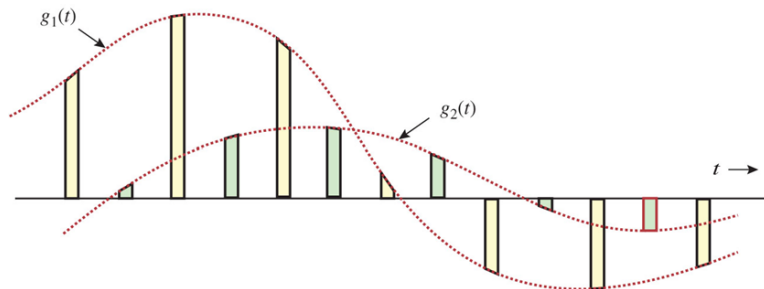
$$B_T =$$

PULSE MODULATIONS

- **PAM:** primarily now used for baseband digital communications (e.g., USB4 Ver 2.0 uses PAM-3, or 3-level PAM)
- **PWM:** power or motor control (e.g., robotics, power supply regulation); minimal telecommunication applications
- **PPM:** fiber optical communications, deep-space communications, R/C applications



TIME DIVISION MULTIPLEXING



BINARY PCM - EXAMPLE PROBLEM 5.2

- $m(t)$ band-limited to 3kHz is sampled at 1/3 higher than the Nyquist rate
- Max acceptable quantization error is 0.5% m_p
- Find min channel bandwidth for transmission.
- If 24 signals combined via TDM, what is min required transmission bandwidth of multiplexed signal?